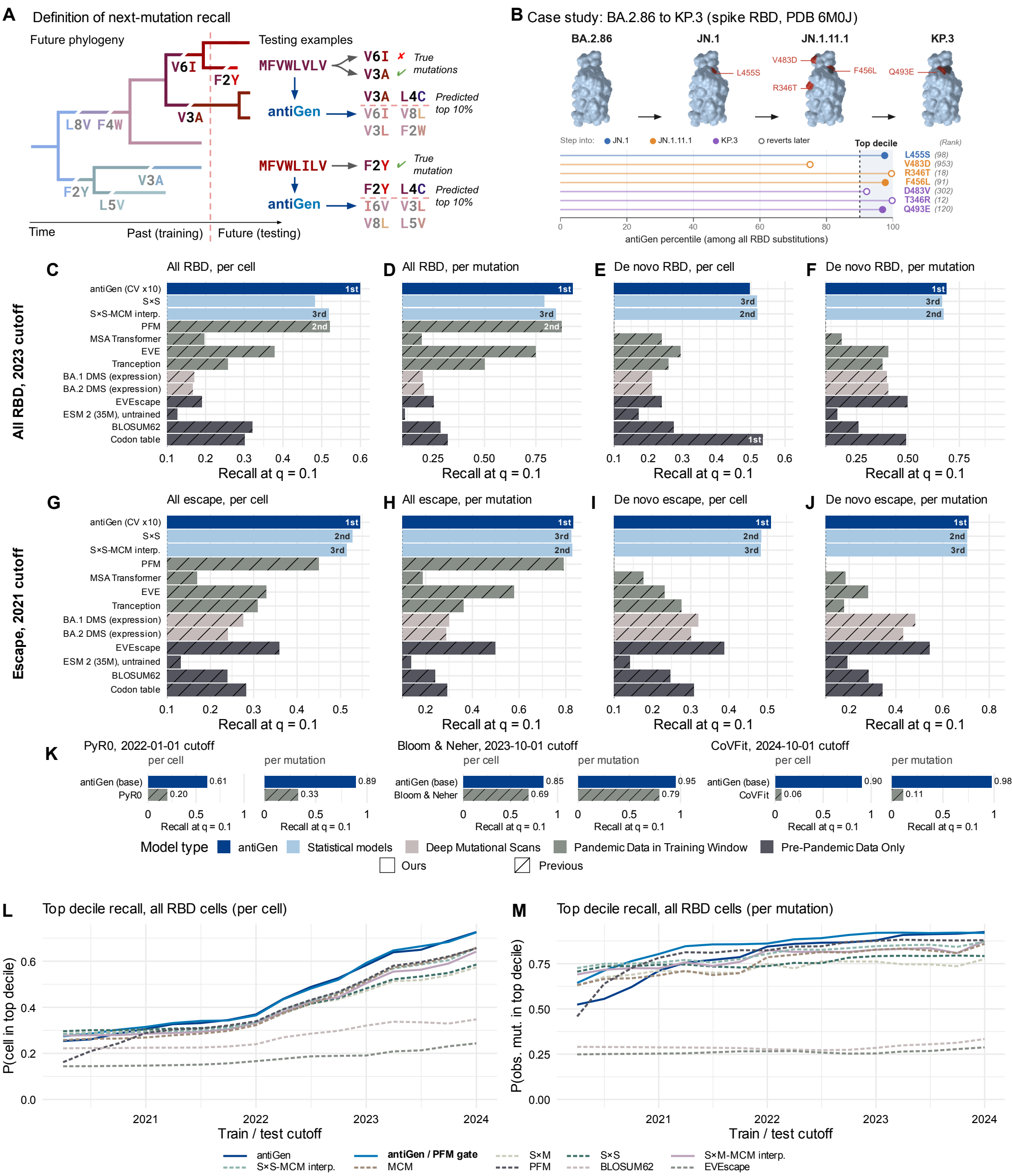

A
Definition of next-mutation recall
Future phylogeny
Testing examples
MFVWLVLV
V6I
V3A
True mutations
antiGen
V3A L4C
V6I V8L
V3L F2W
Predicted top 10%
MFVWLILV
F2Y
True mutation
antiGen
F2Y L4C
I6V V3L
V8L L5V
Predicted top 10%
Time
Past (training)
Future (testing)
B
Case study: BA.2.86 to KP.3 (spike RBD, PDB 6M0J)
BA.2.86
JN.1
JN.1.11.1
KP.3
L455S
V483D
F456L
R346T
Q493E
Step into:
JN.1
JN.1.11.1
KP.3
reverts later
Top decile
(Rank)
L455S (98)
V483D (953)
R346T (18)
F456L (91)
D483V (302)
T346R (12)
Q493E (120)
antiGen percentile (among all RBD substitutions)
C
All RBD, per cell
D
All RBD, per mutation
E
De novo RBD, per cell
F
De novo RBD, per mutation
All RBD, 2023 cutoff
antiGen (CV x10)
S×S
S×S-MCM interp.
PFM
MSA Transformer
EVE
Tranception
BA.1 DMS (expression)
BA.2 DMS (expression)
EVEscape
ESM 2 (35M), untrained
BLOSUM62
Codon table
Recall at q = 0.1
G
All escape, per cell
H
All escape, per mutation
I
De novo escape, per cell
J
De novo escape, per mutation
2021 cutoff
K
PyR0, 2022-01-01 cutoff
Bloom & Neher, 2023-10-01 cutoff
CoVFit, 2024-10-01 cutoff
per cell
per mutation
antiGen (base)
PyR0
Bloom & Neher
CoVFit
Model type
antiGen
Statistical models
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
L
Top decile recall, all RBD cells (per cell)
M
Top decile recall, all RBD cells (per mutation)
P(cell in top decile)
P(obs. mut. in top decile)
Train / test cutoff
antiGen
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
MCM
PFM
BLOSUM62
EVEscape